Anatolii Vakaryuk

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Projects

2026-Present **Quantitative Finance Bootcamp, Erdős Institute**

- Employed probabilistic techniques to forecast stock movement volatility using historical data.
- Evaluated portfolio value at risk and priced European options through advanced Monte-Carlo simulations.
- Enhanced simulation accuracy with control variates, comparing results to Black-Scholes solutions.
- Developed hedging strategies to minimize risk and improve profit distributions in option investments.

Employment

August 2025 - Present **Teaching Assistant, Binghamton University**

- Preparing teaching materials for undergraduate courses in introductory statistics and probability.
- Creating, formatting, and grading quizzes to ensure clarity and consistency for students.

October 2025 - July 2026 **Research in Fourier Analysis, Binghamton University**

- Reverse-engineered the shape of drum heads from the sound waves they create using Laplace eigenfunctions and Fourier analysis.

July 2025 - August 2025 **Camp Counselor, Epsilon Camp**

- Supervised precocious preteens during lectures on algebraic geometry and Turing machines, for an annual math camp hosted at Washington University in St. Louis.

November 2022 - May 2023 **Directed Reading Program, Stony Brook University**

- Studied the proof of the Banach - Tarski paradox under the supervision of PhD student Austin Konkel.
- Presented truncated, bite-sized proof to participants and supervisors at the end of the program.

May 2023 - August 2023 **Knot Theory Research, Stony Brook University**

- Read, studied, and solved problems from *Knot Theory* by Charles Livingston under the guidance of professor Olga Plamenevskaya.

Education

Binghamton University GPA 3.85

Ph.D. Mathematics, 2025–Present

M.A. Mathematics, Expected December 2026

Stony Brook University & Binghamton University GPA 3.75 (Combined)

B.S. Mathematics, 2021–2025

Skills

- Technical Skills: Java, Python, JavaScript, HTML, LaTeX, Numpy, Matplotlib
- Languages: English (Native), Ukrainian (Native)

Publications and Preprints

(Not) hearing where certain triangular drums are struck
arXiv preprint arXiv:2607.10556 (2026)